

# Amol Tripathi

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## Education

Vellore Institute of Technology, Bhopal

Bachelor of Technology in Computer Science and Engineering

Oct 2022–Jun 2026

GPA: 8.34/10.0

## Technical Skills

- **Programming Languages:** C++, Python
- **Web technologies:** HTML, CSS, React.js, Next.js
- **Backend & APIs:** APIs, Node.js, Express, Django, Flask
- **Databases:** MySQL, MongoDB, Vector Databases (Weaviate, ChromaDB)
- **DevOps & Version Control:** AWS (EC2, S3, RDS), Docker, CI/CD Pipelines, Git/GitHub
- **AI/ML Engineering:** RAG Systems, NLP, LLM Integration, Multi-Agent Systems, Prompt Engineering

## Projects

**CodeGenie – AI-Powered Coding Assistant** (Python, LangGraph, LLM, Groq API)

[Link](#)

- Architected a multi-agent AI system using LangGraph to simulate a full software development lifecycle, assigning distinct roles (Planner, Architect, Coder) for autonomous code generation.
- Integrated Groq API for high-speed inference, enabling the transformation of natural language prompts into fully functional applications (e.g., REST APIs, Single Page Applications).
- Optimized agent workflows to automate task decomposition and file generation, reducing boilerplate coding time by approximately 70% for standard web projects.

**FoodieBot- Food Ordering Chatbot** (FastAPI, MySQL, Dialogflow, Python, NLP)

[Link](#)

- Engineered a scalable NLP chatbot capable of handling 200+ concurrent test orders with a sub-2-second latency, deploying the backend on FastAPI.
- Designed a normalized MySQL database schema with stored procedures to ensure ACID compliance and 100% data consistency across order tracking and price calculations.
- Enabled the order management workflow by integrating Dialogflow intents with the backend, reducing manual processing effort by 60%.

**Resume Builder – SaaS Platform** (React, Node.js, MongoDB, HTML, CSS, JavaScript)

[Link](#)

- Developed a full-stack resume generation platform serving 100+ users, featuring streamlined template rendering and real-time preview capabilities.
- Implemented PDF generation logic and optimized frontend state management, streamlining the user journey and cutting resume creation time by 50%.
- Enhanced User Experience (UX) through a responsive interface, resulting in a 40% increase in successful document exports.

## Academic Achievements and Certifications

- Top 5 finalist out of 150+ teams in a college-level hackathon, solving real-world issues in 24 hours.
- Oracle Certified AI Foundations Associate - Recognized for expertise in AI foundations
- Codeforces Specialist (Rating: 1532) demonstrating strong algorithmic thinking and competitive programming skills and consistently solved diverse DSA problems under time constraints.
- Solved 300+ DSA problems (LeetCode, GFG)

## Leadership & Co-curricular

- **Project Team Lead:** Garbage Detection Model – Led a team to develop a YOLO-based computer vision model for roadside garbage detection, improving waste classification accuracy and enabling automated monitoring.
- **Core Committee Member:** In college fests, managed logistics for 1000+ attendees

## Additional Information

- **Languages:** Hindi (Native), English (Fluent)
- **Interests:** Competitive Programming, Financial Markets, Innovation